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Reconstruction of downburst wind fields using physics-informed neural network

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ABSTRACT

Downbursts, as a strong localized wind event, have caused significant damage to engineering structures throughout the world. However, given the spatial and temporal randomness of such strong winds, on-site measurements are often difficult to obtain a sufficient amount of valid wind field information in a short period of time. To refine the resolution of the wind field, this study proposes a physics-informed neural network network-based (PINN-based) approach to reconstruct the downburst from limited observed data. The Navier-Stokes (N-S) equations are embedded into the fully connected neural network as a physical constraint to construct the PINN. The PINN model is then validated by the reconstruction of numerical downburst generated by large eddy simulations. The reconstruction of the sparse downburst wind field by PINN performs well in both interpolation and extrapolation prediction. The optimal construction of the PINN has been evaluated through parameter analysis of the influence of training data and network parameters. Finally, the optimal PINN construction is used to reconstruct the wind field of the experimental data with a relative error of 5% for the horizontal wind velocity.

1. Introduction

Inspired by the strange starburst pattern of forest damage, Fujita (1976) first postulated the existence of "downbursts", defined as strong downdrafts inducing damaging wind bursts near the ground. Based on extensive observations from three research projects, Fujita has defined the radial outflow and divergent winds phenomenon generated by strong downdrafts after touching the ground as a "downburst". Downbursts occur more frequently than tornadoes (Solari et al., 2015), and the probability of a microburst during a thunderstorm is as high as 60–70% (Proctor, 1988).Although most downbursts have a life cycle of 5~10 min (Wilson et al., 1984), their extreme horizontal wind velocity can reach 67 m/s at an altitude of 4.9 m near the ground (Fujita, 1990), posing a great threat to the safety of engineering structures. Downburst outflow winds have been the cause of damage to buildings and infrastructures in many parts of the world (Loredo-Souza et al., 2019). More than 80% of weather-related transmission line failures worldwide are caused by downburst events (McCarthy and Melsness, 1996) and more than 90% in Australia (Li, 2000). Recognizing that downburst has such a severe destructive capacity, research on downburst wind fields has been widely conducted.

Over the past 50 years, a range of well-known full-scale field measurements have been conducted, including: Northern Illinois Meteorological Research (NIMROD) (Fujita, 1979); Joint Airport Weather Studies (JAWS) (McCarthy et al., 1982); The Classify, Locate, and Avoid Wind Shear (CLAWS) (McCarthy, 1985); the Microburst and Severe Thunderstorm (MIST) (Fujita, 1990); Wind and Ports (WP) (Solari et al., 2012); the Severe Convective Outflow in Thunderstorms (SCOUT) (Gunter and Schroeder, 2015); Wind, Ports and Sea (WPS) (Repetto et al., 2017); ERC THUNDERR Project (Solari, 2020). These observational studies provide valuable data on the formation, evolution, and wind field characteristics of downbursts. Based on the 6-year wind data of 6 meteorological stations in Hong Kong, Shu et al., (2015) discussed the influencing factors of the gust factor in downbursts, noting that the deviation of the gust factor is related to terrain conditions and wind velocity. After analyzing the measured data of a typical thunderstorm event in the mountainous area of western China, Huang et al. (2019) pointed out that the measured thunderstorm wind characteristics are not significantly different from those in the flat area. Loredo-Souza et al. (2019) performed a parametric analysis of two specific downburst cases

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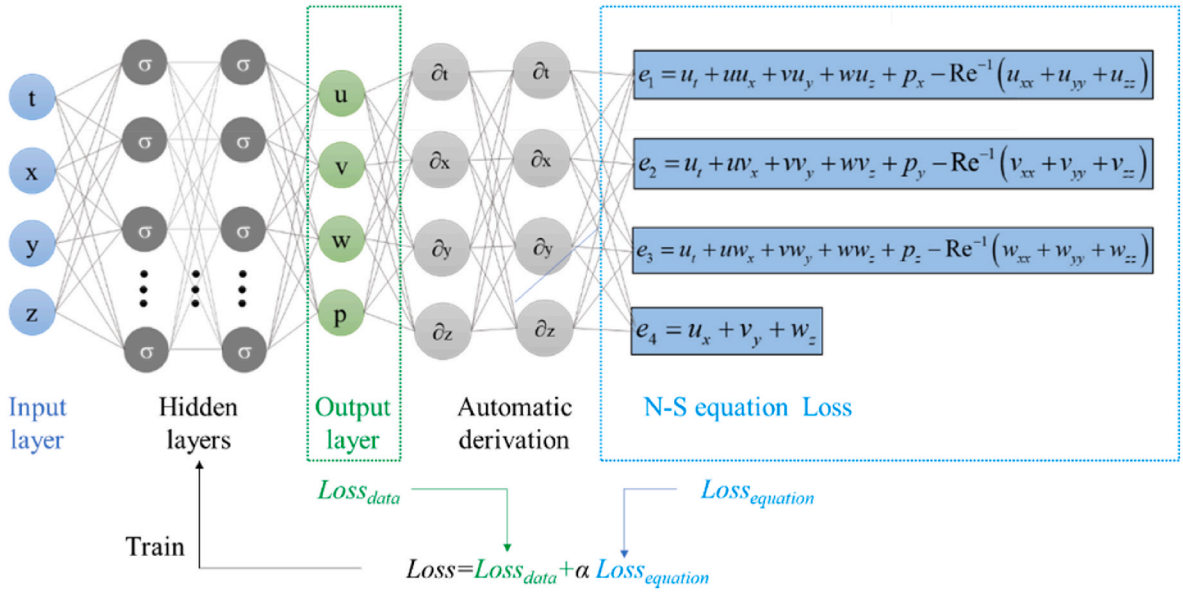


Fig. 1. Framework of the PINN, u, v and w correspond to the velocity in the x, y, and z directions, respectively. First-order derivative $u_x = \frac{\partial u}{\partial x}$, second-order derivative $u_{xx} = \frac{\partial u_x}{\partial x}$, the same goes for the rest of the derivatives. Re is Reynolds number , p denotes the fluid pressure.

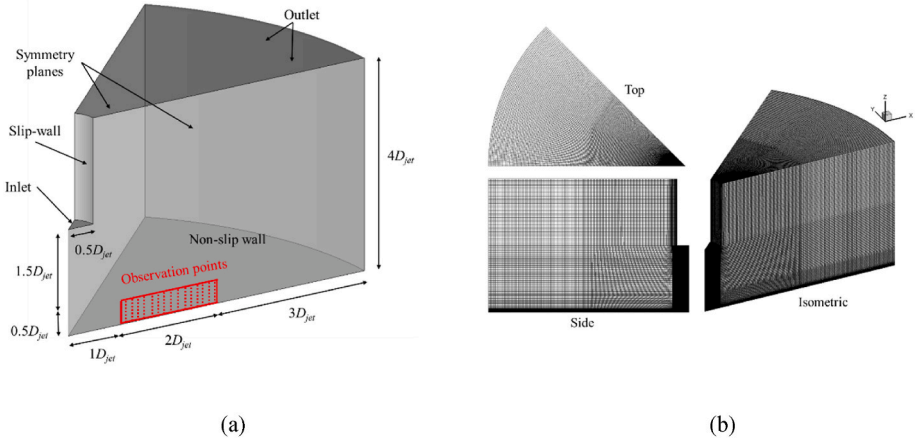


Fig. 2. (a) Compute domains and boundary conditions and (b) the mesh system.

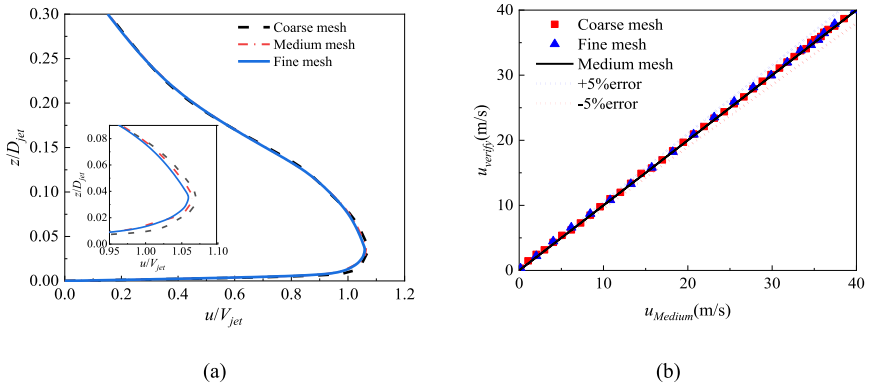


Fig. 3. (a) Vertical horizontal wind profile and(b) mesh error verification.

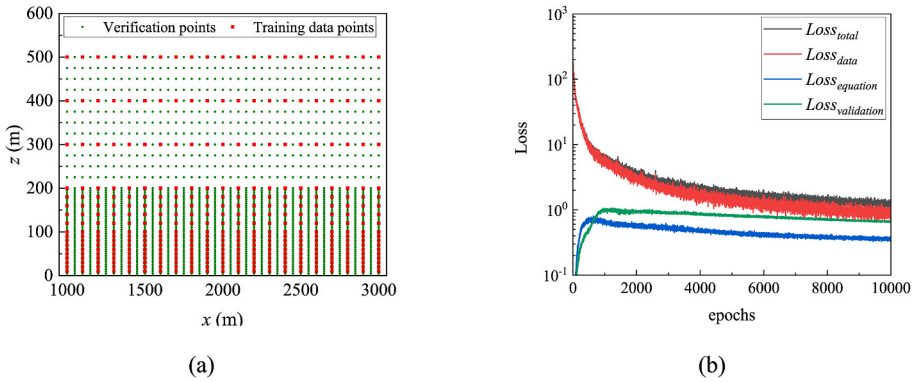


Fig. 4. (a)The distribution of training measurement points and (b)the loss function decreased.

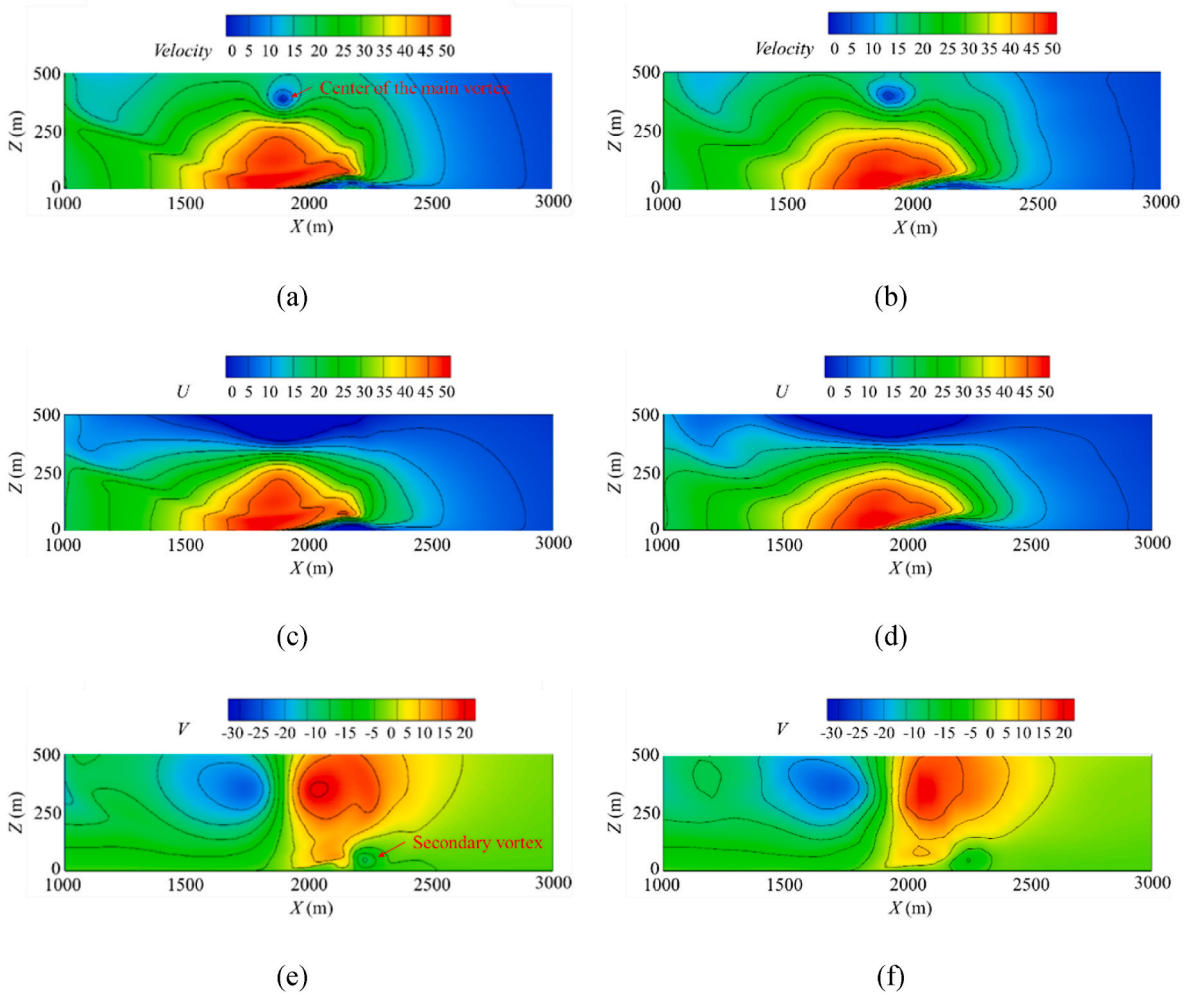


Fig. 5. The velocity contours at $t = 200$s (a) CFD V_{max} (b) PINN V_{max} (c) CFD u (d) PINN u (e) CFD v (f) PINN v, where u is the horizontal velocity component, v is the vertical velocity component, and V_{max} is the vector synthesis velocity.

that occurred in Brazil and analyzed the failure patterns of downbursts on building structures. Based on the continuous wind and structural response monitoring of three slender structures in the THUNDERR project, Mengistu et al. (2023) analyzed the wind-induced response of the monitored structures in two downburst cases. Although field measurements of full-scale downbursts are often considered one of the most reliable research methods, the randomness of occurrence, small scale of wind impact, strong destructiveness, and unpredictable path make it difficult and expensive to measure extreme winds such as downbursts (Kim and Matsui, 2017). At the same time, the data obtained from downburst measurements are often inadequate and of low resolution, which is not sufficient for direct application to structural wind engineering. Therefore, it is necessary to reconstruct the sparse flow field of downburst with high resolution to obtain high-resolution wind field data.

Physics-informative neural network (PINN) is a novel neural network framework that combines data-driven and physical law constraints (Raissi et al., 2019), which has become an important method for

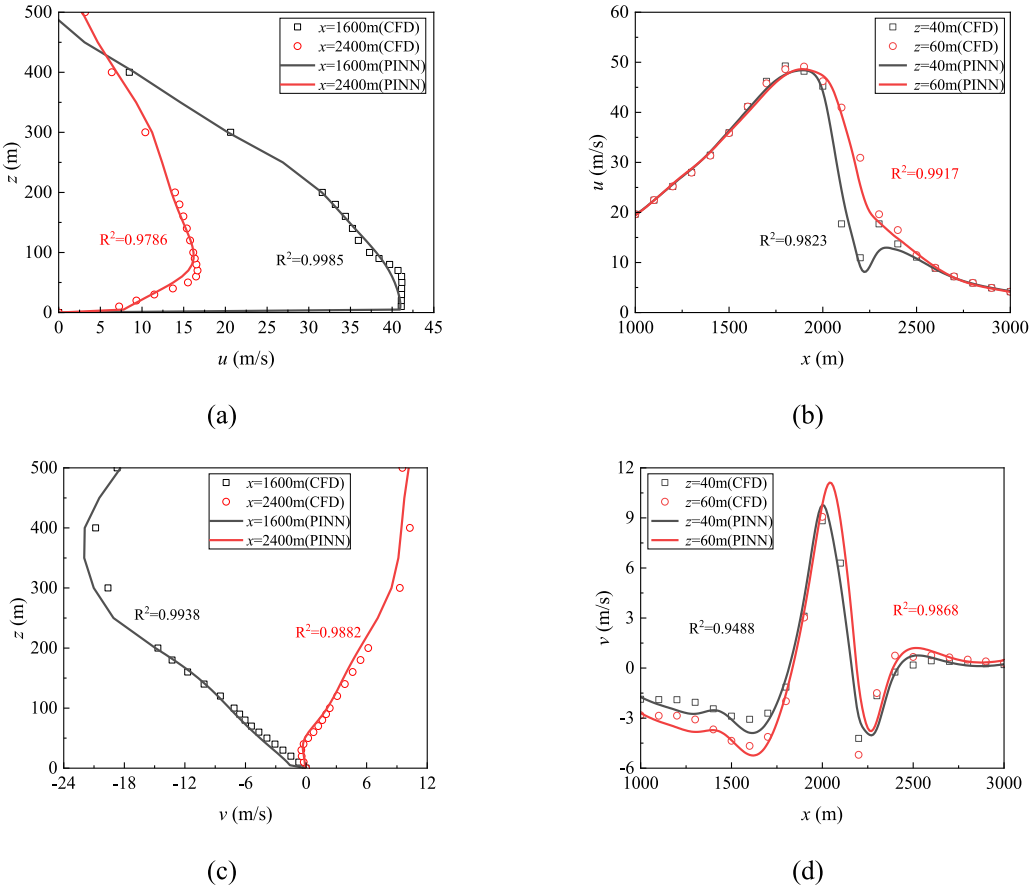


Fig. 6. Wind velocity profiles verification: (a) u vertical (b) u radial (c) v vertical (d)v radial.

solving machine learning problems with small data samples. The introduction of physical partial differential equations makes the predictions as consistent as possible with the laws of physics (Raissi et al., 2020). Although PINN is still a work-in-progress in terms of solving flow problems without any given data (Chuang and Barba, 2022). However, based on its ability to reconstruct the flow field of sparse measurement points, PINN have been tried in areas such as vortex induced vibration (Bai and Zhang, 2022; Jin et al., 2021; Raissi et al., 2018; Yang and Mesri, 2022), medical imaging (Karniadakis et al., 2021), fluid heat transfer(Cai et al., 2021; Heger et al., 2024), component transport (Laubscher, 2021; Tucny et al., 2024), particle-laden flows (Seyed-Ahmadi and Wachs, 2022), and physical experiments of fluids (Wang et al., 2022) and exhibited excellent flow field resolution improvement capabilities. In wind engineering research, Zhang et al. (2023) used limited observational data to reconstruct tornado vortices, and the relative error of reconstruction was less than 10%. However, the reconstructed tornado wind field is a steady-state wind field. Eusebi et al. (2023) used a sparse set of real-time observations to accurately reconstruct Hurricane Ida. The introduction of the PINN method provides a promising alternative for data assimilation for tropical cyclones initial conditions to significantly reduce the computational time. PINN is much more efficient than CFD in terms of memory resource usage, with PINN requiring 5–10 times less memory (Ang et al., 2023).

In this study, we proposed a PINN-based method to reconstruct the transient full-size downburst wind field using a small amount of flow field data, which solves the problem of limited data and low resolution of the downburst wind field in the wind resistance analysis of civil structures. Section 2 introduces the PINN methodology used in this study. In section 3, a transient simulation of unsteady downburst is performed, and then the PINN is tried to reconstruct the downburst to verify its feasibility. Then, through parameter analysis, this study evaluates the ability of the PINN network to reconstruct the downburst field and searches for the optimal PINN setup. Section 4 applies the optimal PINN to reconstruct the downburst wind field measured in the wind tunnel test. Conclusions are presented in section 5.

2. Physics-informed neural network

A physics-informed neural network (PINN) is a type of deep neural network (DNN). Classical DNN is a kind of data-driven method, which means that the prediction is determined by the quality and quantity of the training data. Conversely, embedding the physical equations into the loss function of a traditional DNN greatly reduces the training data dependence of PINN. In this study, the Mach number of the downburst near-surface flow is considerably less than 1.0. As a result, the governing physical equations embedded in the loss function are based on the classical incompressible Navier-Stokes (N-S) equations. Fig. 1 illustrates the framework of the PINN, which primarily comprises input layers, hidden layers, and output layers.

Cartesian spatial coordinates (x, y, z) and time t are inputted into the input layer, and velocity and pressure are predicted through fully connected neurons in the hidden layer. Adaptive Swish(Wang et al., 2022) is chosen as the activation function because it has the best performance in the reduction of training loss and the smallest difference between training loss and validation loss.The adaptive swish function σ is expressed by:

$$\sigma(x) = \frac{x}{1 + e^{-n\beta x}} \tag{1}$$

where n represents a scale factor, β is a trainable parameter. A larger n

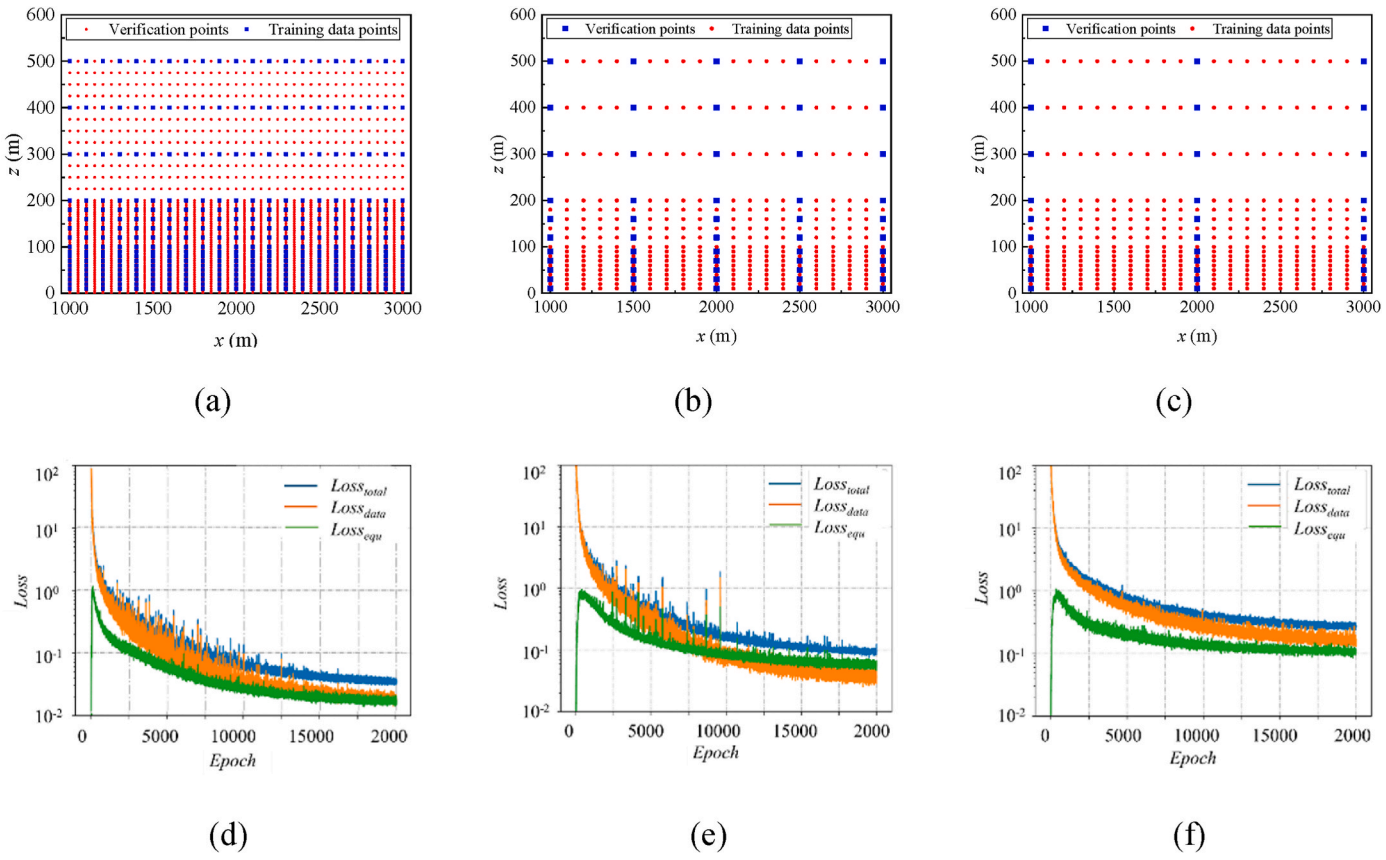


Fig. 7. The distribution of measurement points (a) 378 (b) 55 (c) 33 and their loss functions decreased (d) 378 (e) 55 (f) 33.

can accelerate the convergence procedure. Therefore, the scale factor n is set to 10, and the initial value of β is set to 0.1(Jagtap et al., 2020).The partial differentiation of the prediction results is obtained by automatic differentiation in TensorFlow 2.1.

The PINN's loss function is comprised of two components: the difference between the predicted and observed values at observation points and the residual of the embedded physical equations. The weight of each component is determined by the weighting coefficient α. The loss function of a PINN is defined as follows:

$$Loss = Loss_{data} + \alpha Loss_{\text{equations}}$$

$$Loss_{data} = \frac{1}{N_{data}} \sum_{i=1}^{n} \left[\left(u_{pred}^{i} - u_{true}^{i} \right)^2 + \left(v_{pred}^{i} - v_{true}^{i} \right)^2 + \left(w_{pred}^{i} - w_{true}^{i} \right)^2 \right]$$

$$Loss_{\text{equations}} = e_1 + e_2 + e_3 + e_4 \tag{2}$$

where N_{data} denotes the numbers of measured data involved in the training, u_{pred}^{i} , v_{pred}^{i} and w_{pred}^{i} indicate the prediction results of the PINN. On the other hand, u_{true}^{i} , v_{true}^{i} and w_{true}^{i} represent the true values at the observation points.

The optimizer utilized is Adam(Kingma, 2015), which relies on first- and second-order moment adaptive estimation in stochastic gradient descent. The training process is complete when the loss function is less than the target value or when the number of training steps reaches a specified number of epochs.

3. Reconstruction of numerical simulation downburst wind field

Based on the PINN framework constructed in Section 2.1, numerical simulation of the downburst was carried out and the finite flow field data measured is used to reconstruct the main vortex of the downburst to provide high-resolution information about the wind field near the surface of the downburst.

3.1. Numerical simulation of unsteady downburst

In this section, ANSYS FLUENT was used to simulate a numerical downburst using an unsteady impinging jet model. The computational domain, as depicted in Fig. 2a, where the outflow diameter of the simulated downburst, denoted as D_{jet}, is specified as 1 km. The speed and turbulence intensity of the inlet are 30 m/s and 1%, respectively. Consequently, the geometric scale ratio is maintained at a 1:1 proportion. To guarantee the full development of the downburst and the stability of the boundary conditions as outlined by Sengupta and Sarkar (2008), the calculation domain's length and height were set at $6D_{jet}$ and $4D_{jet}$, respectively. The impinging jet inlet has an intake velocity of 30 m/s and a turbulence of 1%, which corresponds well to real meteorological conditions, so the velocity scale ratio is also 1:1, with the time scale ratio being 1:1 accordingly. The Reynolds stress model (RSM) is used for numerical simulation, the governing equations are solved using the separation method for state simulation, the time discretization is in the second-order implicit scheme, the pressure-velocity coupling equation is in SIMPLEC algorithm, and the momentum, TKE, TDR and Reynolds stress are in QUICK scheme. A total of 378 observation points were used to collect the time history information of the velocity within 400s with a time step of 0.1s, which will be used for PINN training and the verification of prediction results.

The computational domain is divided by a structured hexahedral mesh to ensure computational efficiency and accuracy, as shown in Fig. 2b. The height of the first layer cell is set to 2×10^{-5} m, and the growth rate along the height is 1.1 to meet the dimensionless distance requirement (Haines and Taylor, 2019). The vertical horizontal wind profile at 1.6 km from the impact center was selected to verify the mesh

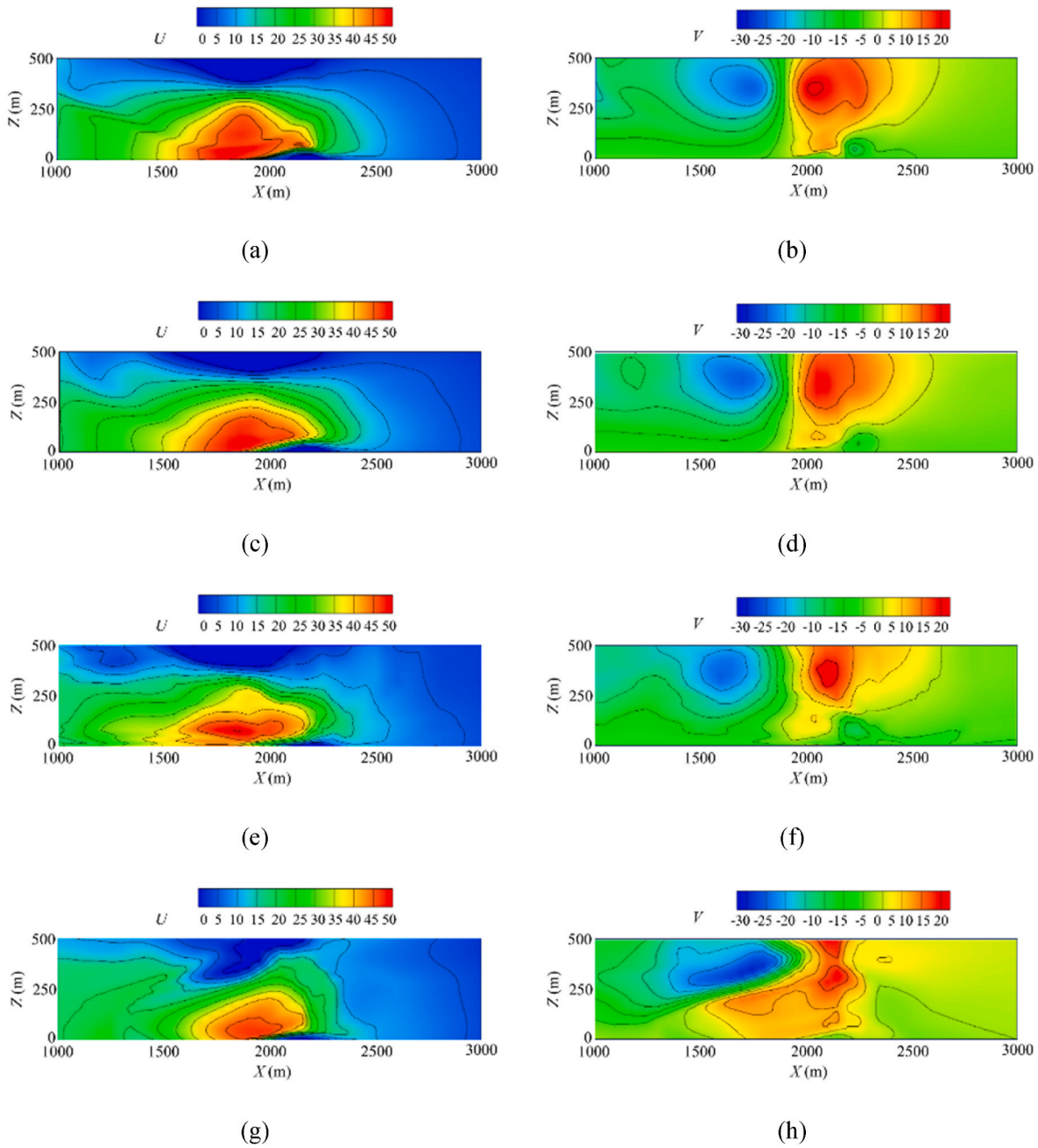


Fig. 8. The influence of observation point spacing on the prediction results of contour (a) CFD u (b) CFD v (c)378 u (d)378 v (e)55 u (f)55 v (g)33 u (h)33 v.

independence, as shown in Fig. 3. The results of the three mesh systems are generally not more than 5% different, but there are significant differences in the near-ground position of the wind profile. Therefore, the medium mesh system is adopted to conduct the simulation to ensure the accuracy of results and computational efficiency.

3.2. Performance of the PINN approach to reconstruct downburst

As suggested by Raissi, a network architecture with 7 hidden layers and 32 neurons per layer was used to reconstruct the downburst flow field (Raissi et al., 2019). The weighting factor a is set to one. The training data is about 1.5×10^4 samples, which is 1% randomly selected from 4001 time series data for velocities u and v of 378 observation points. The number of equation training points is 1×10^4, which are randomly distributed in the training space. The adaptive moment estimation (Adam) optimizer was used to train 1×10^4 epochs with an initial learning rate of 1×10^{-3}. The learning rate decays exponentially in a staircase manner (Wang et al., 2022).The decline of the loss function is shown in Fig. 4, which decreases rapidly at the beginning of training and gradually stabilizes as the training epochs increase. The total loss function is mainly composed of the loss of training data, while the loss of the equation is relatively small. The verification error is stable at about 3.1% at the end of the training.

Since the velocity component w is negligible compared to the horizontal velocity u and vertical velocity v on the selected radial slice, the reconstruction of velocity u and v is the primary focus of this demonstration. As illustrated in Fig. 5, the first main vortex structure of the downburst is adequately developed at $t = 200$s, and an extreme wind velocity region is formed at this location. The vortex center of the main vortex is located near $x = 1.8$ km and $z = 400$m. For the distribution of horizontal wind velocity, the extreme wind velocity appears below the vortex center, and there is a stationary region with zero velocity on the

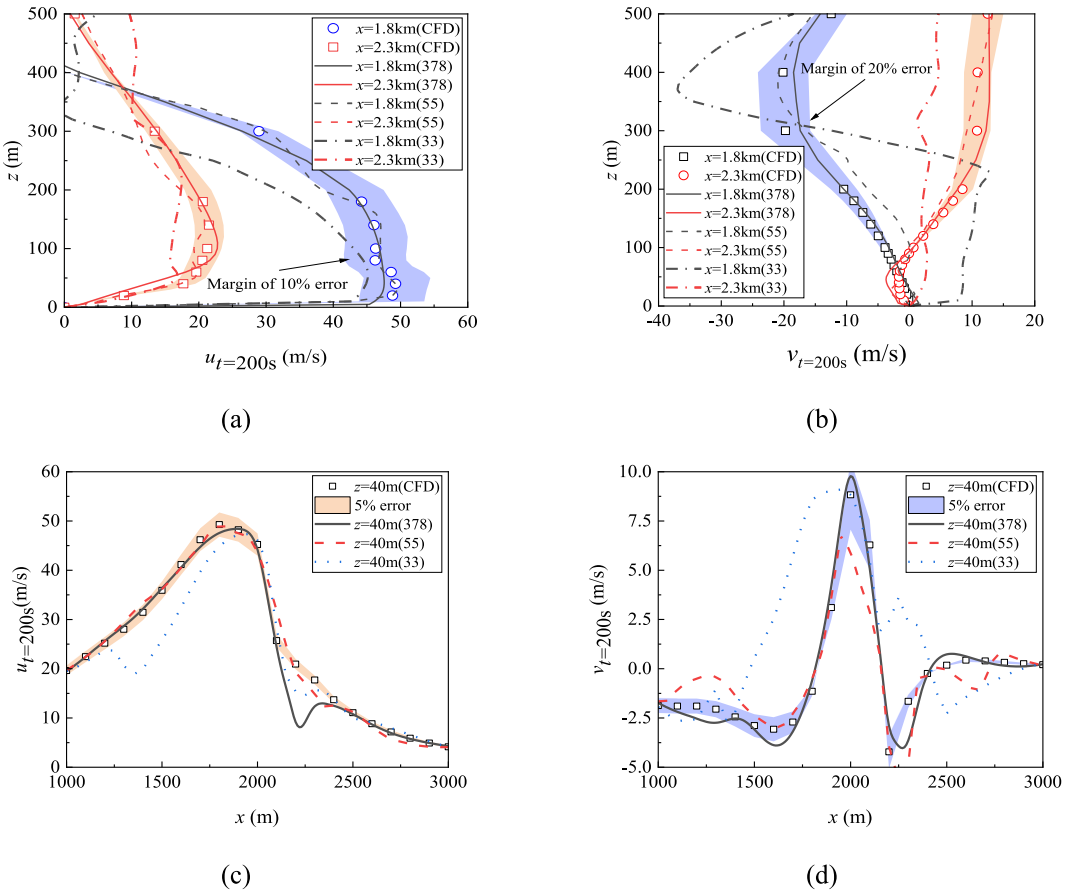


Fig. 9. The influence of observation point spacing on the prediction results of profile (a) vertical u (b) vertical v (c) radial u (d) radial v.

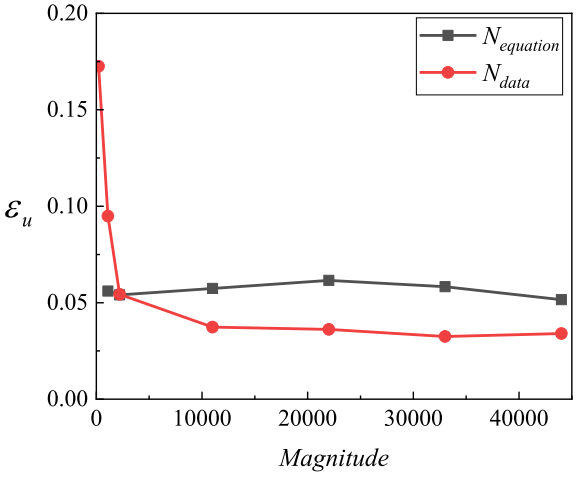


Fig. 10. The effect of the number of training points on the reconstruction error.

ground before the extreme wind velocity region. The vertical velocity contour is a diagram of two opposite velocity distributions tangent at the vortex center. PINN can accurately reconstruct complex flows in the vicinity of the main vortex, especially the magnitude and location of extreme wind velocity, and the center of the vortex is also displayed. The prediction results of the PINN below the center of the vortex underestimate the wind velocity due to the increase in the vertical spacing of the measurement points in this region. For the velocity contour of the vertical wind velocity, it can be found that the PINN has a good representation of the two typical opposite velocity distributions of the vertical wind velocity on both sides of the main vortex, and even the secondary vortex near the ground at the front end of the main vortex is clearly depicted in the contour of the vertical velocity v.

As shown in Fig. 6a, the typical nose-shaped wind profile is well displayed at $t = 200$s, and the horizontal wind velocity at $x = 1.6$ km is considerably faster than that at $x = 2.4$ km. The results at $x = 1.6$ km and $x = 2.4$ km demonstrate a high degree of concordance with the CFD results, with fitting correlation coefficients R^2 for the two are 0.9985 and 0.9786, respectively. For the radial wind profile at $t = 200$s (Fig. 6b), there is a distinct peak around $x = 1.6$ km, which corresponds to the maximum horizontal wind velocity at the location of the main vortex. PINN's reconstruction of the radial wind profile coincides perfectly with the CFD results. The verification positions of the vertical wind profile are located on both sides of the main vortex, so the wind velocity direction is opposite (Fig. 6c). The radial wind profile of the vertical wind velocity changes twice with the increase of the radial distance, and the position of the change is the vortex center position (Fig. 6d).

3.3. Parametric study

The selection of the PINN network structure and the amount of training data have a direct impact on the prediction accuracy of PINN. This section employs the same CFD data utilized in the previous section to examine the spacing of the observation points, the matching of the neural network structure, and the amount of training data. This serves as a reference for the subsequent reconstruction of the impinging jet test data.

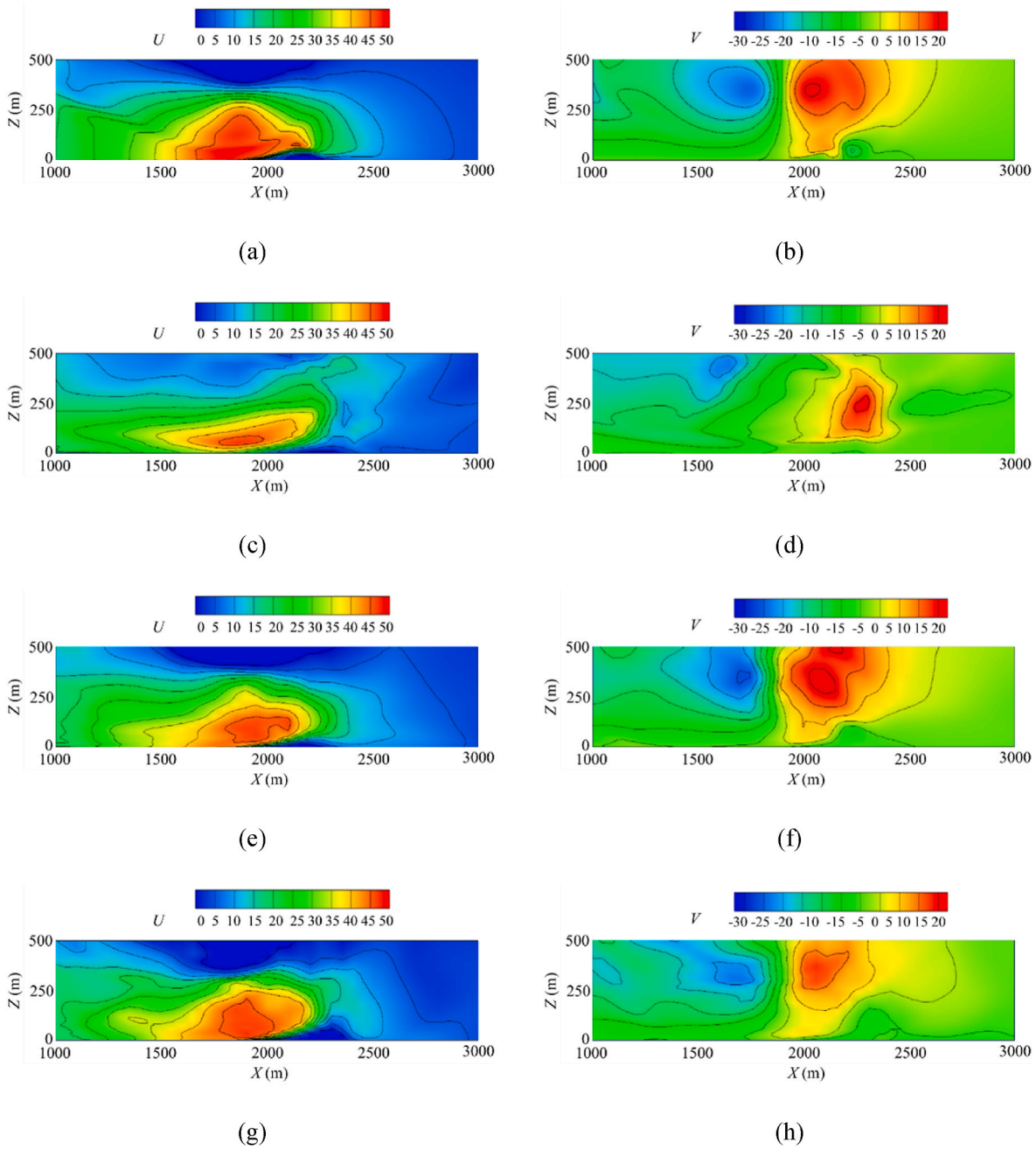


Fig. 11. Velocity contour under different N_{data}: (a) CFD u (b) CFD v (c) $N_{data} = 0.1\%$ u (d) $N_{data} = 0.1\%$ v (e) $N_{data} = 5\%$ u (f) $N_{data} = 5\%$ v (g) $N_{data} = 20\%$ u (h) $N_{data} = 20\%$ v.

In contrast to traditional neural networks, which necessitate a substantial quantity of training data to achieve satisfactory prediction outcomes, PINN requires a markedly smaller amount of training data and physical equation constraints to attain high-precision flow field prediction. This section examines the impact of observation point spacing and the quantity of data on the prediction results of PINN, with the objective of achieving an acceptable level of prediction accuracy with the minimum amount of training data.

In addition to the 378 observation points at 100m spacing utilized in the previous analysis, the number of observation points is further reduced. The number of observation points was further reduced by selecting 55 points at 500m spacing and 33 points at 1000m spacing, respectively (Fig. 7). Based on the above three types of observation points, the results of the PINN reconstruction of the downburst wind field are compared. To eliminate the potential influence of other variables, the network structure uses the same 7 (layers) × 32 (neurons), the number of training data is 1.5×10^4, the equation points are kept at 1×10^4, and a training epoch of 2×10^4 iterations. All the other parameters are the same. As the distance between measurement points increases, the final stable value of the loss function also increases. This also implies a reduction in the accuracy of the PINN model.

As illustrated in Fig. 8, the velocity contour predicted by PINN with dense observations is the most closely aligned with the CFD results, indicating that the prediction is the most accurate. As the spacing between observation points increases, the prediction of the complex flow of the downburst becomes increasingly distorted, particularly in the case of the contour of the vertical velocity v. As the spacing between observation points increases, the correlation between the data points involved in the training weakens, which makes it more challenging for the PINN to reconstruct the complex flow of the downburst. Although the velocity

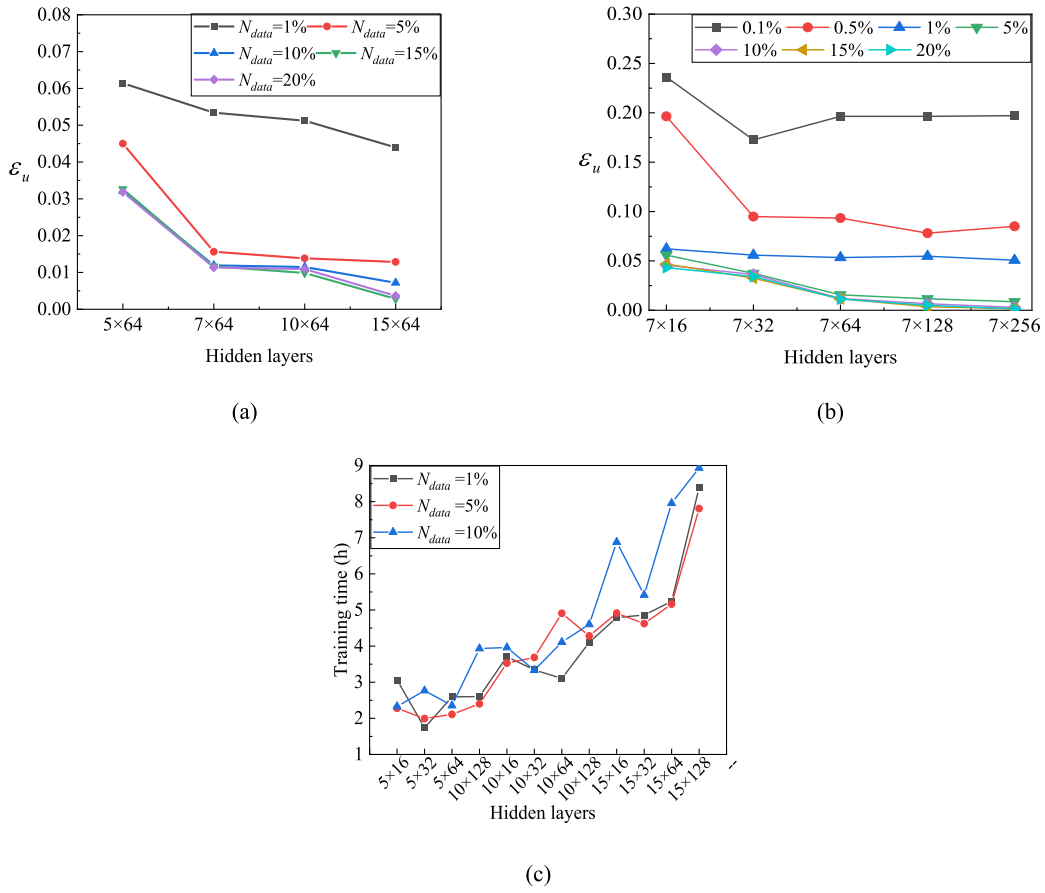


Fig. 12. The influence of network structure and training data on the prediction results of PINN (a) hidden layer depth, (b) number of neurons in the layers and (c) training time for different network structures.

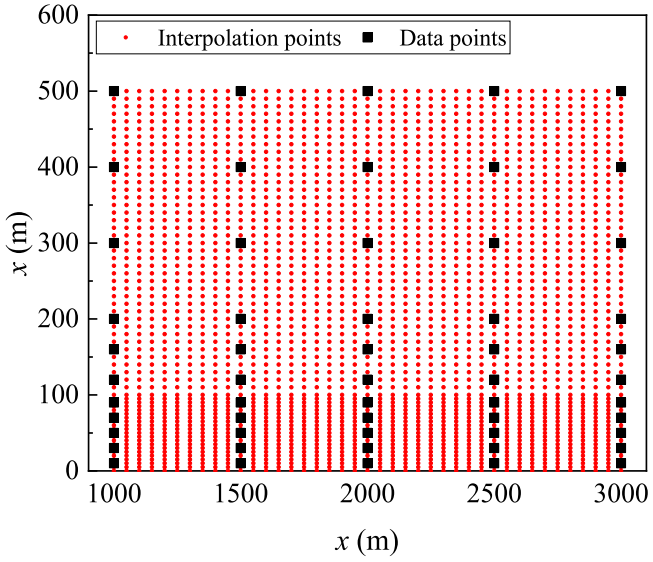


Fig. 13. PINN interpolation training and validation points.

contours predicted by the 55 observation points are not as accurate as those of the 378 observation points, they are still acceptable in terms of the near-surface horizontal wind velocity and the main vortex structure. However, the velocity contours predicted by PINN at the 33 observation points are significantly distorted, and the vertical velocity contour result is the least accurate. Consequently, the subsequent parameter studies will utilize the 500m spaced observation points.

As Fig. 9 shows that as the horizontal spacing of measurement points increases, the accuracy of PINN prediction of wind profiles will decrease significantly. The results of 378 and 55 are considerably superior to those of 33 points, and the influence on the reconstruction accuracy of horizontal wind velocity is less than that for vertical wind velocity. When the horizontal spacing is less than 500m, the reconstruction error of horizontal wind velocity is less than 10%, while that of vertical wind velocity is less than 20%.

In order to investigate the influence of the amount of data used for training on the prediction results of PINN, different amounts of training data (N_{data}) and equation points ($N_{equation}$) were employed, with all other settings remaining consistent, based on the data of 55 observation points. As illustrated in Fig. 10, the prediction error exhibits a rapid decline at the outset as the Ndata is augmented. However, at $N_{data} = 1 \times 10^4$, the prediction error plateaus, indicating that further increases in N_{data} do not result in a reduction in the prediction error. In comparison to N_{data}, an increase in $N_{equation}$ does not result in a notable reduction in the prediction error.

The velocity contours of PINN trained with different N_{data} at t = 200s are presented in Fig. 11. Upon reaching 1% of the total observed data (2.2×10^3), the predictions of the velocity contours are found to be in general agreement with the results of the CFD simulation. When the N_{data} exceeds 5% (1×10^4), increasing the N_{data} does not result in a notable enhancement in the accuracy of the prediction outcomes, despite maintaining the same network structure. This is corroborated by the reconstruction error results presented in Fig. 10. As the value of N_{data} increases, the prediction of the vertical velocity v contour becomes

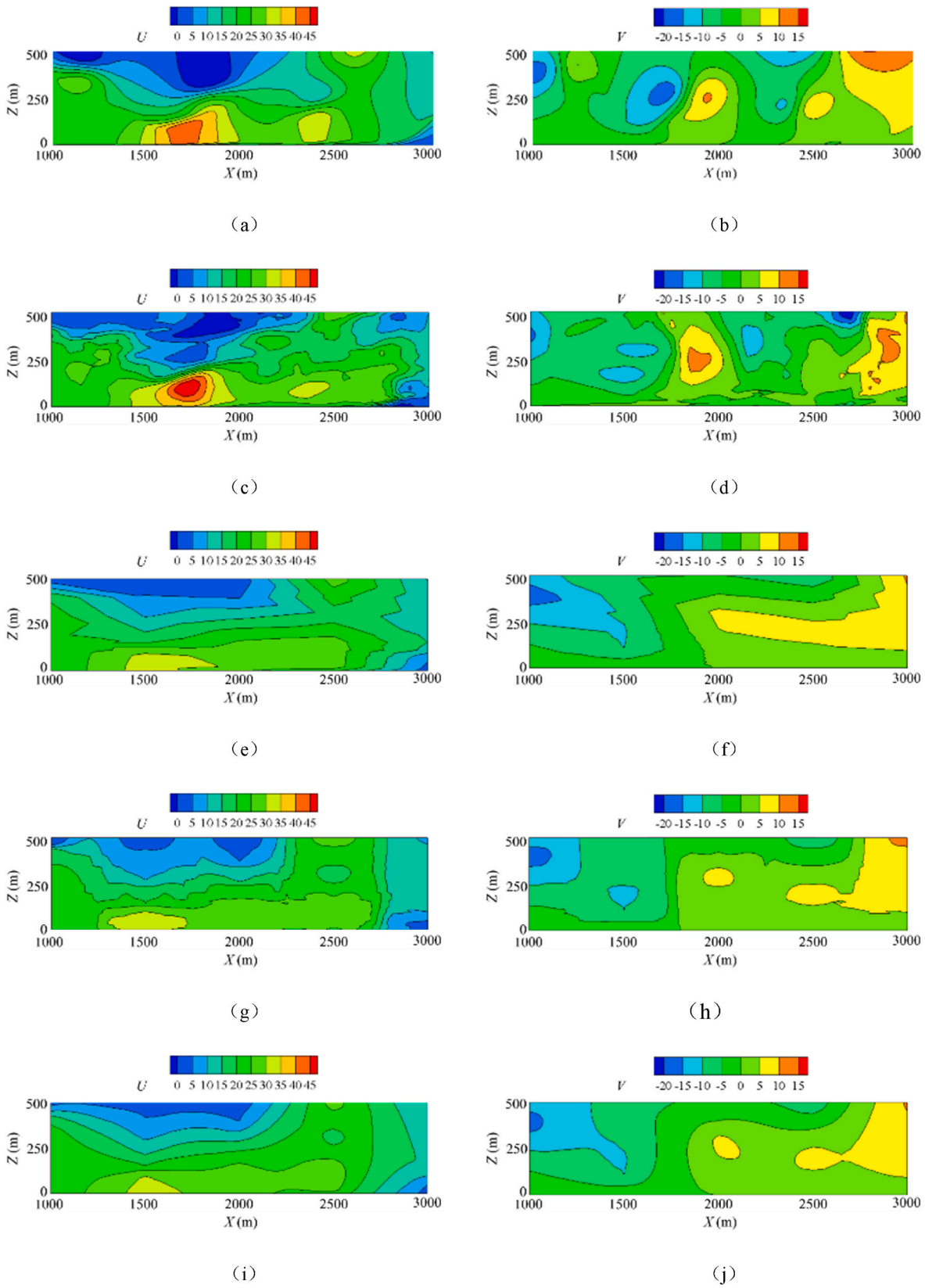


Fig. 14. The (a) CFD u (b) CFD v (c) PINN u (d) PINN v (e) linear interpolation u (f) linear interpolation v (g) inverse distance interpolation u (h) inverse distance interpolation v (g) kriging interpolation u (k) kriging interpolation v.

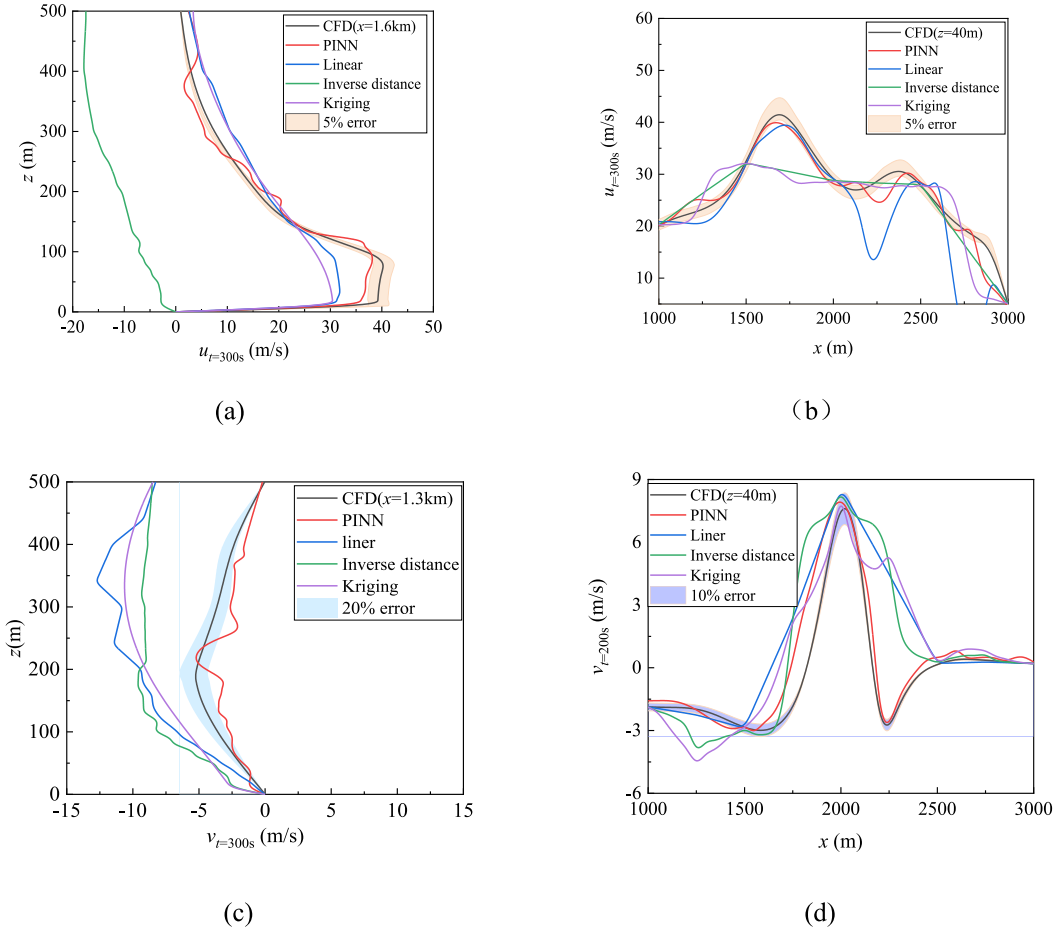


Fig. 15. Comparison of interpolated radial wind profiles (a) x = 1.6 km t = 300s u (b) z = 40 m t = 300s u (c) x = 1.3 km t = 300s v (d) z = 40 m t = 200s v.

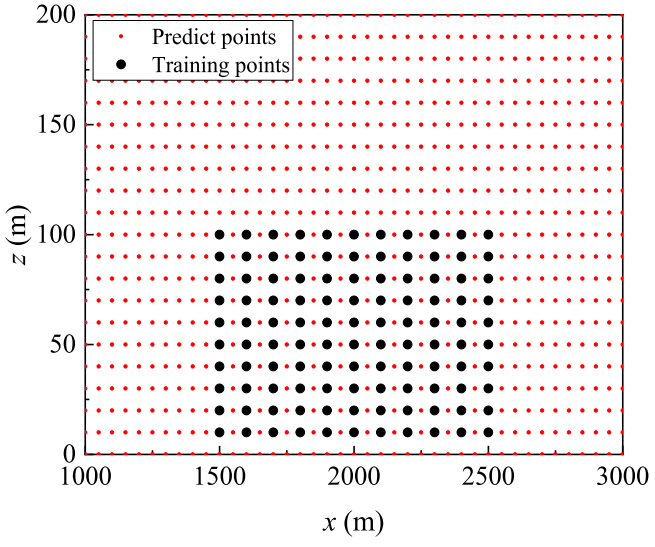


Fig. 16. Distribution of trained data points and extrapolation points.

increasingly smooth, indicating the presence of an overfitting phenomenon.

It can be postulated that an increase in the number of hidden layers and neurons within a network may enhance its ability to predict high-dimensional nonlinear problems. This would result in more accurate predictions for complex flows. However, an increase in the number of hidden layers and hidden neurons also entails a significant expansion in the number of trainable parameters, resulting in a proportional increase in the time required to train the network. To investigate the influence of PINN network structure on the prediction results of PINN, a series of experiments were conducted using N_{data} values of 55 observation points. These experiments involved reconstructing the downburst wind fields using different PINN network structures.

The prediction error for horizontal wind velocity as predicted by the PINN is illustrated in Fig. 12. As the N_{data} increases, the prediction error decreases significantly. However, when the quantity of training data is insufficient, the augmentation of the number of layers and neurons within the hidden neural network does not result in an enhancement of the prediction accuracy. Upon reaching a data volume of greater than 5% (1×10^4), the discernible enhancement in prediction accuracy resulting from an increase in the N_{data} becomes less pronounced. The training time of different network structures is illustrated in Fig. 12c, as the complexity of the network structure increases, so does the time required to train the PINN. Therefore, for the reconstruction of the downburst wind field, the network structure of 7×32 and $N_{data} = 5\%$ (1×10^4) can be used to obtain optimal prediction results at a low computational cost.

3.4. PINN interpolation predictions

A total of 5% of the flow field data from 55 measurement points (as illustrated in Fig. 13) were selected as the training data, and the network was trained for 2×10^4 epochs with a 7×64 network structure. To

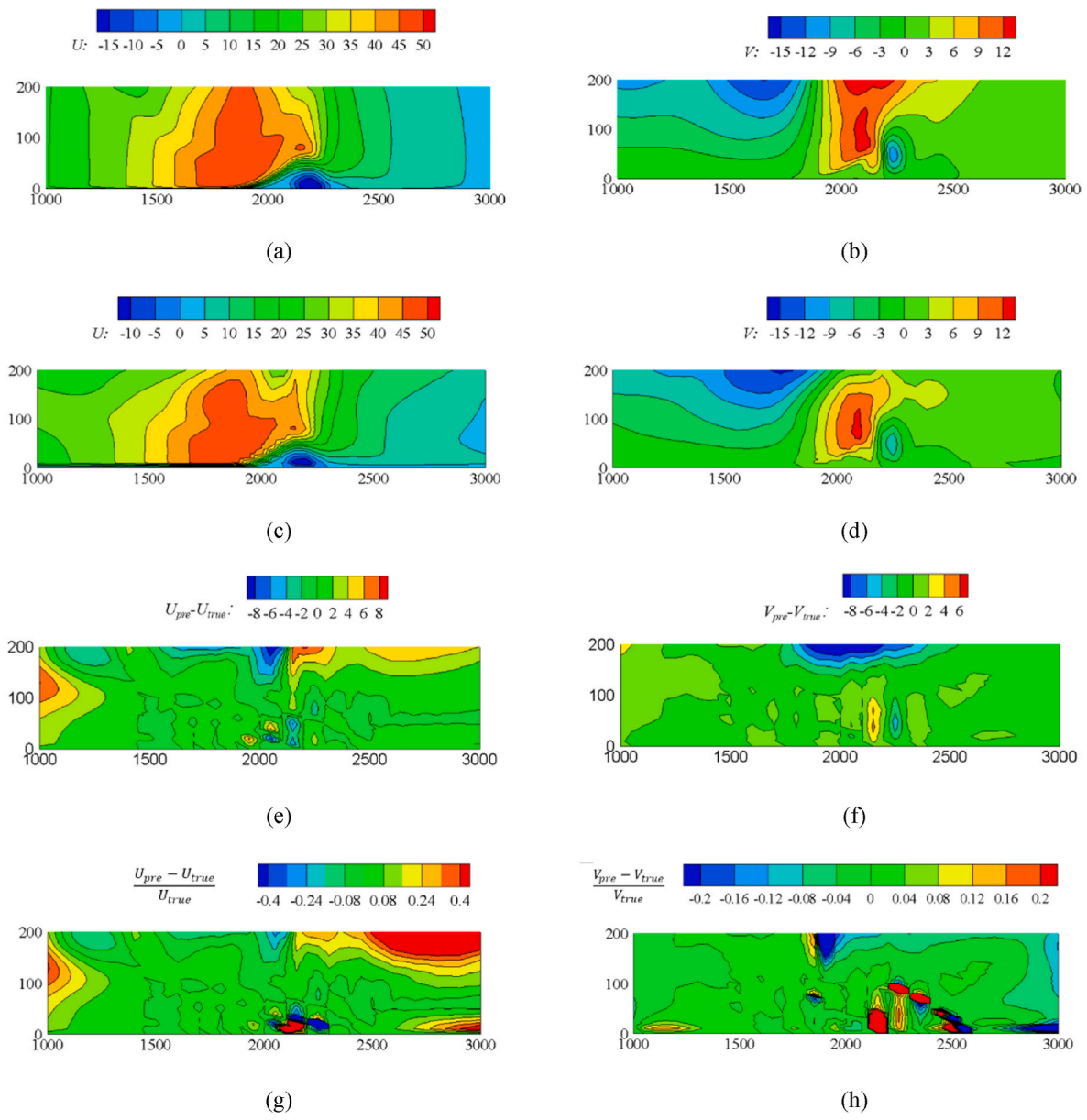


Fig. 17. PINN extrapolation prediction velocity contours (a) CFD u (b) CFD v (c) PINN u (d) PINN v (e) absolute error of u (f) absolute error of v (g) relative error of u (h) relative error of v.

assess the interpolation accuracy of PINN relative to that of interpolation methods, the selected interpolation methods include linear interpolation, inverse distance interpolation and kriging interpolation.

As illustrated in Fig. 14, the superiority of PINN over traditional interpolation methods is even more pronounced at $t = 300$s, especially for extreme wind velocity distributions between $x = 1.5$ km and 2 km. The conventional interpolation methods are based on the flow field data at the time of measurement and utilize interpolation algorithms to predict the velocity at an unknown location. However, these methods are not suited when the distances between data points are large, and the data nonlinearity is significant. The PINN not only refers to the measurement point data at the current time, but also participates in the training process using the data from both the preceding and subsequent moments through the flow governing equations. This enables the PINN to achieve high prediction accuracy even in the absence of an unknown measurement point.

As shown in Fig. 15, the PINN prediction error of horizontal wind profile is maintained at a level of 5% or below, while the error of other interpolation methods is significantly higher. At $x = 1.6$ km, the primary advantage of PINN is its enhanced accuracy in predicting wind velocity near the surface. The radial distribution of horizontal velocity in the PINN accurately fits the CFD results, whereas the traditional interpolation method significantly underestimates the velocity spike between $x =$ 1.5 km and $x = 2$ km. In the case of vertical wind velocity, while the prediction results of PINN are superior to those of traditional interpolation methods, the error of PINN reaches 20%.

3.5. PINN extrapolation predictions

It has been proved that PINN have better interpolation prediction accuracy than typical interpolation methods under the same conditions, especially when extreme wind velocity occurs in the location of no measurement point distribution. This section will examine the capacity of PINN to address extrapolation prediction issues that are not adequately addressed by traditional interpolation methods.

The distribution of training points and validation points extrapolated

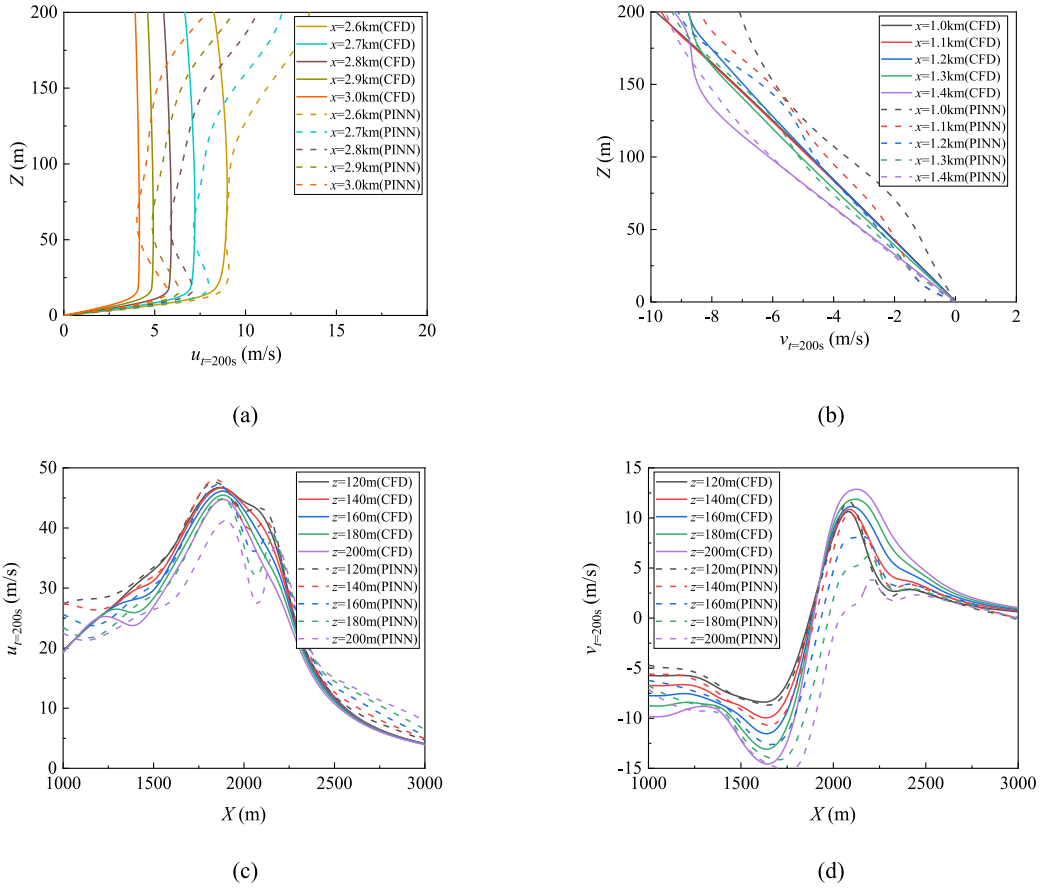


Fig. 18. PINN extrapolated predicted wind profiles: (a) vertical u (b) vertical v (e) radial u (f) radial v.

by PINN is illustrated in Fig. 16. The upstream and downstream and upper positions of the training points are validated by comparing the wind profiles. A total of 5% of the data from 110 training points located at $x = 1.5$–2.5 km and $z = 0$~100m, were employed to train the PINN. The network was trained for 2×10^4 epochs with a 7×64 network structure, with the objective of extrapolating 500m horizontally and 100m vertically.

As shown in Fig. 17, the extrapolation prediction of the horizontal velocity for $t = 200$s produced by the PINN is comparable to the CFD results. At $x = 1.0$ km, the PINN predicts a greater horizontal velocity than is observed in the extrapolation prediction. At $z = 200$m altitude, the horizontal velocity u is overestimated and underestimated in different areas. The absolute error of the PINN's prediction of horizontal wind velocity extrapolation is approximately 8 m/s, while the absolute error of velocity in most areas is approximately 2 m/s. The error percentage contour indicates that the majority of errors are within 10%. However, the excessive relative error is due to the fact that the true velocity of certain positions is close to 0 m/s. In the extrapolation prediction of the vertical velocity contour map, it can be observed that the PINN underestimates the upward vertical velocity around $x = 2.0$ km and $z = 200$m. For other positions, the absolute error of the PINN extrapolation velocity is typically within 2 m/s, with a relative error below 10%.

Fig. 18 presents the PINN extrapolation of $t = 200$s, which is employed to predict the wind profile. It can be observed that the error in the vertical wind profile extrapolation increases with the distance over which the extrapolation is performed. Furthermore, the error at high altitude is more pronounced. The extrapolated prediction of the radial wind profile is in good agreement with the CFD results, except for $z = 200$ m at t = 200s and $z = 180$m and 200 m at t = 200s.

4. Reconstruction of downburst experimental wind field

In the previous section, the optimal PINN model is obtained based on the numerical simulation data. In this section, the experimental data from impinging jet is utilized to further validate the PINN model. Firstly, the wind field of the steady downburst on flat and sloping terrain is simulated by using the impinging jet device. The test was conducted by a fixed nozzle and a constant jet speed, and the data was collected when the wind field was fully developed. Subsequently, the wind field reconstruction is carried out using the optimal PINN to demonstrate its efficacy.

4.1. Experimental setup

In the experimental wind field, the jet nozzle diameter D_{jet} is 0.6m. Based on relevant literatures(Fang et al., 2023; McConville et al., 2009), with a similar nozzle diameter, the geometric scale ratio is approximately 1:2000. The turbulence intensity at the nozzle outlet is almost 0, and the jet velocity V_{jet} is constant at 12 m/s, giving a velocity scale ratio of approximately 1:1 to 1:4, and thus the time scale ratio is about 1:2000 to 1:500. The experimental equipment is depicted in Fig. 19. The horizontal distance between the center of the nozzle and the cornice position of the sloping terrain is $1.5D_{jet}$. The slope length l is 300 mm and the slope height h is 150 mm, which yields a slope angle of 30°. More observation points are distributed near the ground to capture the large velocity gradient. The observation points are distributed vertically in the range of 10 mm–180 mm from the ground, with 15 points per column. For each observation point, the horizontal and vertical wind velocities were recorded using hot-wire anemometers with a sampling frequency of 5000 Hz and a sampling time of 20s. Owing to the multitude of

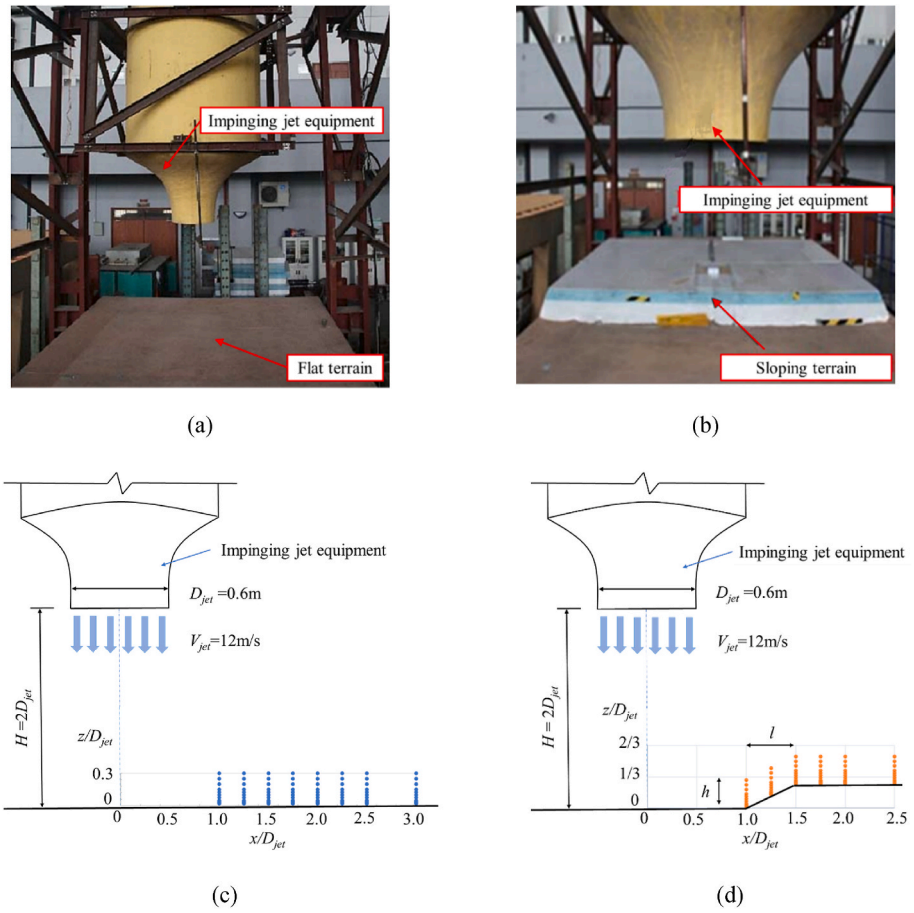


Fig. 19. Photographs of steady impinging jet experiment (a) flat terrain (b) sloping terrain and arrangement of (c) flat terrain measurement points (d) sloping terrain measurement points.

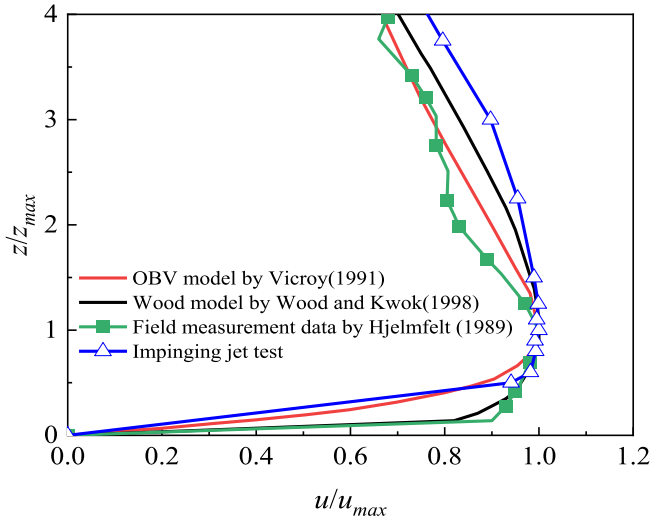


Fig. 20. Wind profiles of the test and related literatures.

measurement points, sampling at each point was executed in a sequential batch process following the attainment of a steady-state wind field. The procedure commenced with the activation of the fans, and subsequent to the stabilization of wind velocities, the collection of near-ground wind speed data was initiated. The experiments used in this study have been detailed in earlier literature(Fang et al., 2019). Fig. 20 shows the comparison of the dimensionless wind profile of the impinging jet test with the results of related literature. Among them, the Wood model (Wood and Kwok, 1998)is in good agreement with the CFD results and the experimental data, while the OBV model (Vicroy, 1991) and field measurement data (Hjelmfelt et al., 1989) deviates slightly.

4.2. *Experimental wind field reconstruction results*

For the flow field reconstruction of wind tunnel test data, a 2D PINN neural network was used. The hidden layer of the network structure adopts 7 layers, 64 neurons in each layer, 5×10^3 equation training points, and 2×10^4 training times. Given that the amount of fluctuating wind velocity data collected by each measurement point reached 1×10^5, 1% of each training measurement point was used as the training sample. As illustrated in Fig. 21, for the reconstruction of the test wind field, half of the measurement points were utilized as the training points, while the remaining half were employed as the verification points.

4.2.1. *Reconstruction of flat terrain wind field*

As Fig. 22 illustrates, the PINN trained with only half of the measurement points performs well in the reconstruction of the wind field on the flat ground, especially for the distribution prediction of horizontal wind velocity. However, the distribution of vertical wind velocity is less accurate. The maximum wind velocity of the downburst is observed at a distance of $1.0D_{jet} = 600$ mm from the jet in the steady state of flat terrain, with a magnitude velocity of approximately 12 m/s. As the radial distance increases, the horizontal velocity decreases gradually, and the high-wind velocity area forms a wedge-like shape in the horizontal wind velocity u contour.

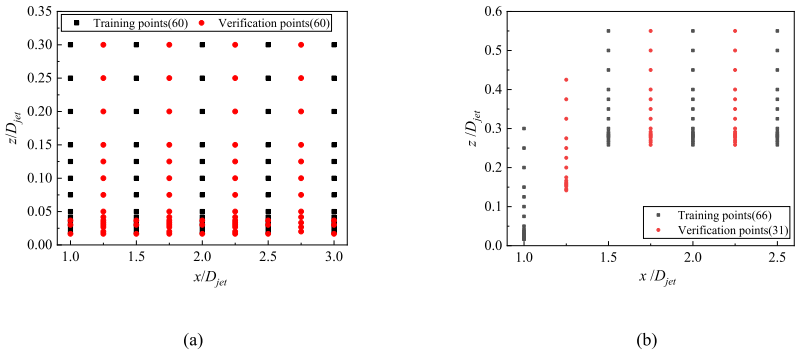


Fig. 21. Training points and validation points of (a) flat terrain and (b) sloping terrain.

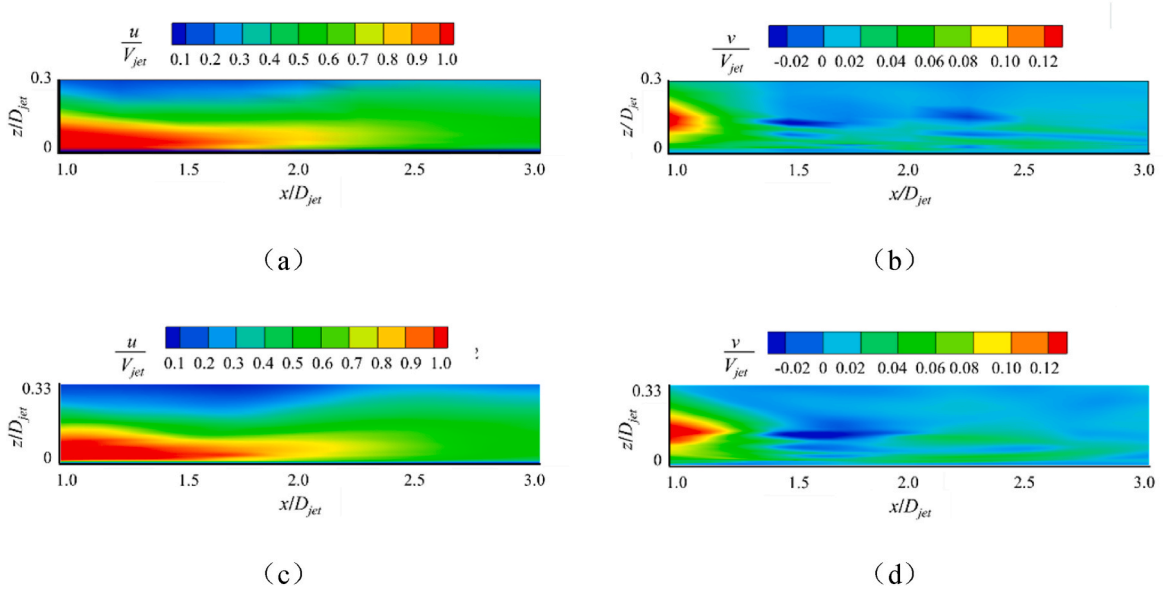


Fig. 22. Flat terrain contours of (a) test u (b) test v (c) PINN u (d) PINN v.

PINN's downburst wind field reconstruction results for flat terrain are presented in Fig. 23. The vertical wind profile of the horizontal wind velocity is characteristic of the nose shape, and the magnitude of the extreme wind velocity decreases with the increase of radial distance. The test results are in good agreement with those of PINN, and the error of PINN prediction is controlled at 5%. From the radial wind profile of the horizontal wind velocity, it can be observed that the horizontal wind velocity decreases with the increase of radial distance and decreases with the increase of altitude. For vertical wind profiles, PINN's prediction of vertical wind velocity is not as accurate as that of horizontal wind velocity, with a deviation of more than 15%. This is due to the relatively low value of the vertical wind velocity, with the maximum velocity being less than 1 m/s. Furthermore, the vertical distribution of the vertical wind velocity exhibits a significant degree of variability. In the verification of vertical wind velocity, the radial distribution generally conforms to the change trend of the experimental data, while the vertical distribution deviates more significantly.

4.2.2. *Reconstruction of slope terrain wind field*

Unlike flat land, which is characterized by a dominant horizontal wind velocity, slope terrain significantly enhances the vertical wind velocity component, as illustrated in Fig. 24. The horizontal velocity at the beginning of the slope and the middle of the slope decreases, while the cornice has a significant acceleration effect. The similarity between the synthetic velocity and the horizontal velocity contour is high, with the vertical velocity mainly appears at the cornice position. The reconstruction of the horizontal wind velocity distribution by PINN is obviously better than that of the vertical wind velocity contour.

The wind velocity profile of the slope terrain is depicted in Fig. 25. The prediction error of the vertical wind profile and the radial wind profile of the horizontal wind velocity is about 5%. Furthermore, the maximum horizontal wind velocity of $x = 1.75D_{jet}$ and $x = 2.25D_{jet}$ at the top of the slope is increased due to the influence of topography. The occurrence of the maximum horizontal wind velocity of $z = 0.03D_{jet}$ is also postponed to the cornice position($x = D_{jet}$) for the radial wind profile of horizontal wind velocity. The predicted value for the vertical wind velocity exhibited a margin of error of 15%.

5. Conclusions

In this paper, a PINN-based method is proposed to reconstruct the downburst from limited and low-resolution observational data. By embedding the classical governing equations N-S equations into the fully connected neural network as physical law constraints, the amount of training data required for the neural network is greatly reduced. The PINN model is validated by reconstructing of the numerical downburst. Then, through the parameter study, the optimal PINN model is sought to provide a reference for the measured downburst wind field

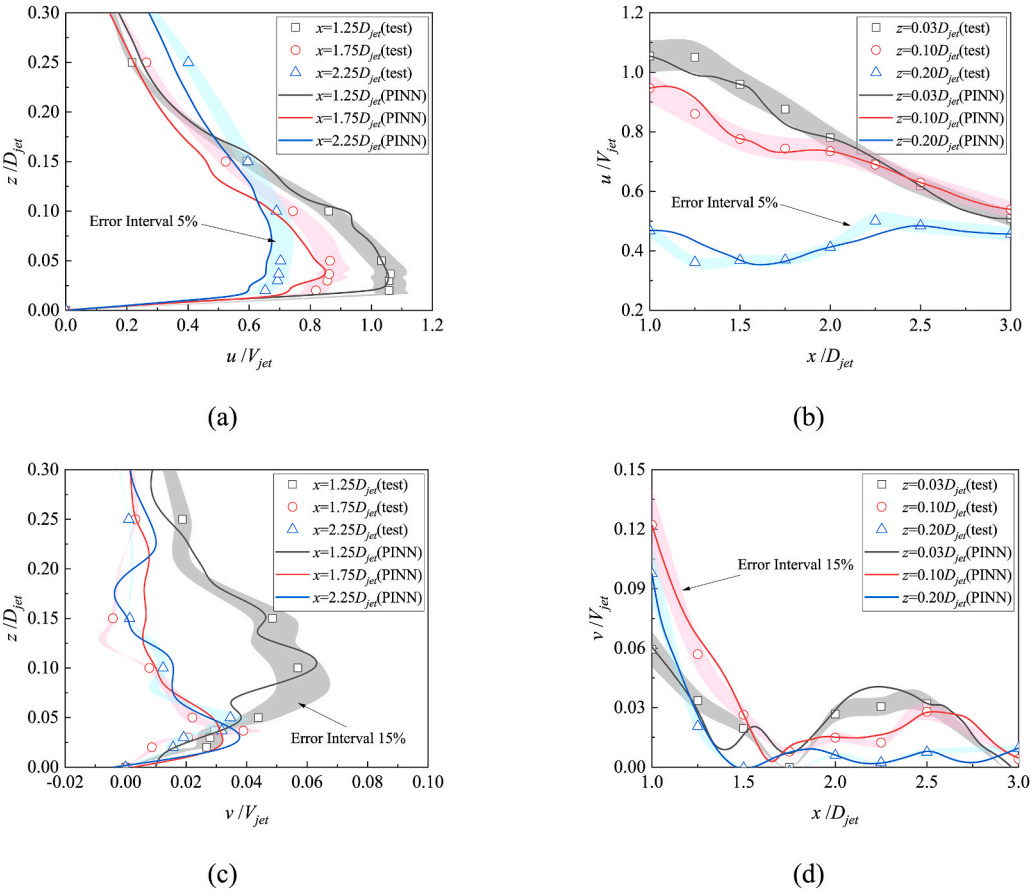


Fig. 23. Wind profile of flat terrain (a) vertical *u* (b) radial *u* (a) vertical *v* (b) radial *v*.

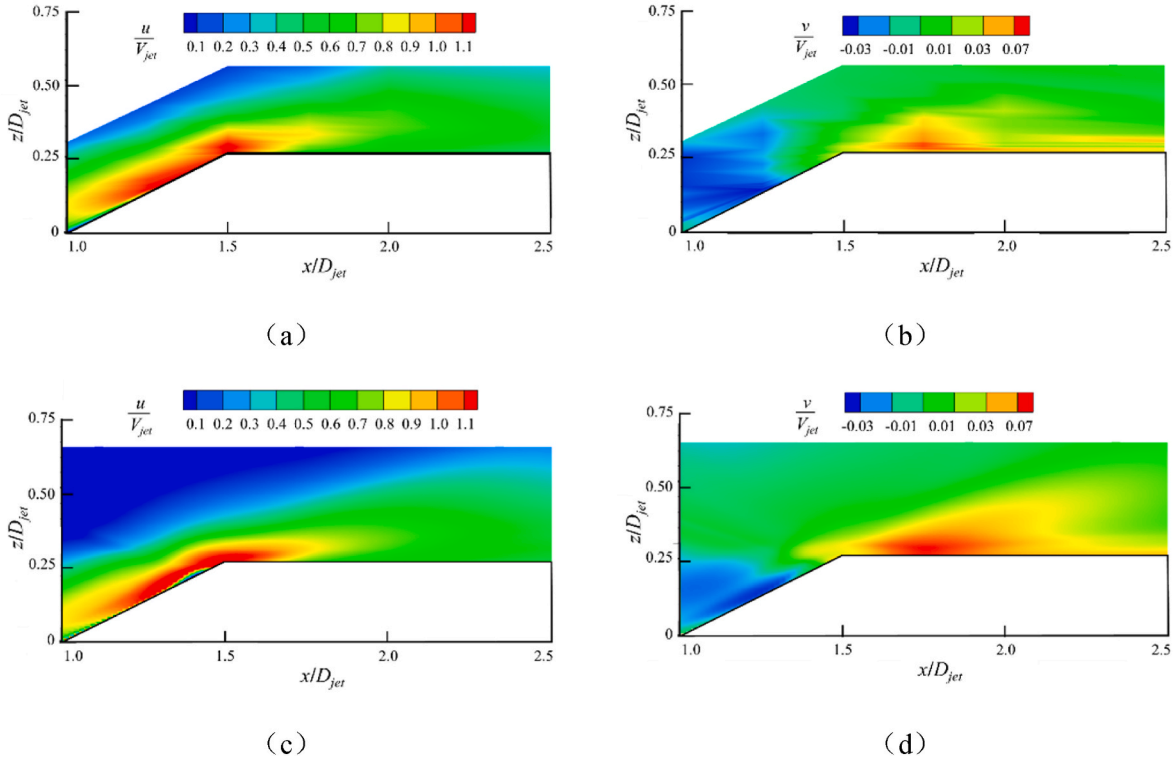


Fig. 24. Sloping terrain contours of (a) test u (b) test v (c) PINN u (d) PINN v.

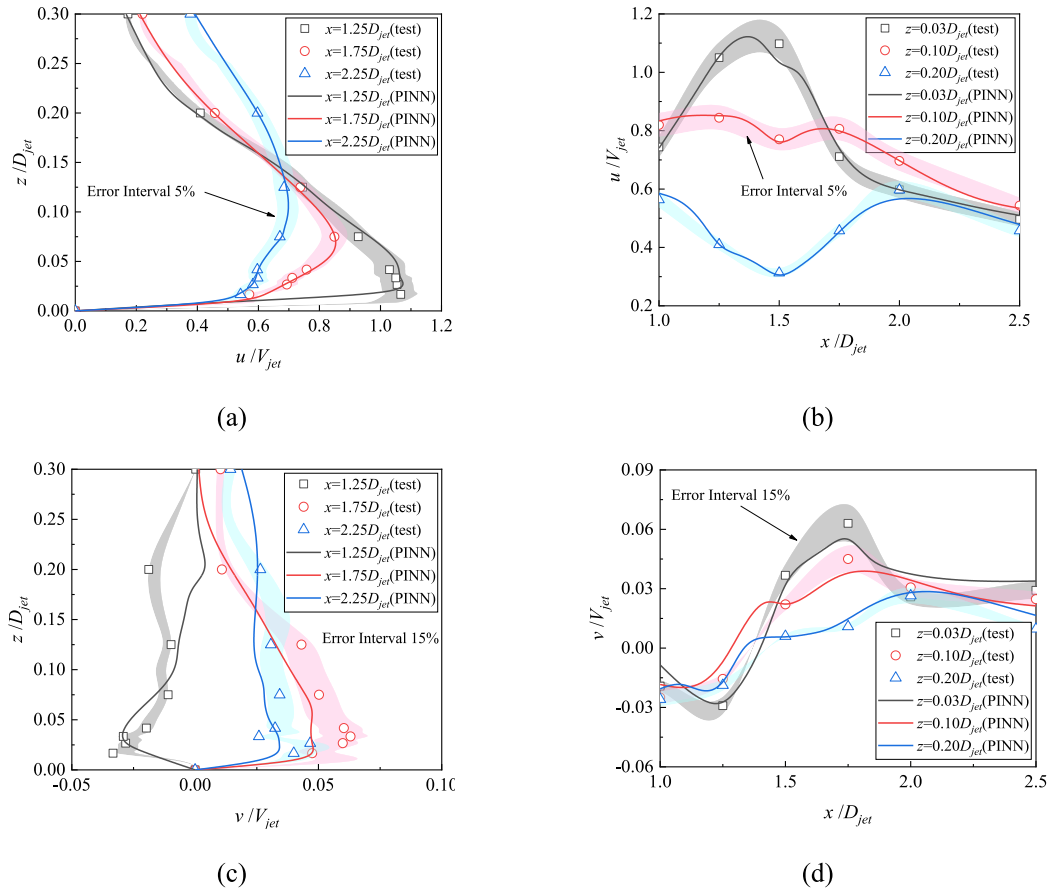


Fig. 25. Wind profile of sloping terrain (a) vertical u (b) radial u (a) vertical v (b) radial v.

reconstruction. Ultimately, the proposed approach is applied to reconstruct the wind field from an impinging jet experiment, thereby substantiating its applicability. The following conclusions are drawn.

(1) The reconstruction of the transient downburst based on PINN can accurately capture the main vortex structure and the extreme wind velocity near the ground. The model's predictions exhibit greater accuracy in horizontal wind velocity estimations compared to those of the vertical component. A comparative analysis with three conventional interpolation methods demonstrates that PINNs possess substantial advantages in interpolating sparse flow field data. Furthermore, PINNs exhibit robust extrapolation capabilities, addressing the challenges in complex flow field predictions that conventional interpolation methods often struggle with.

(2) According to the systematic parametric analysis, the increase in the spacing between observation points leads to a decrease in reconstitution accuracy. Under the same network construction, blindly increasing the amount of data participating in training does not significantly improve the prediction accuracy. Enhancing the network's scale can effectively diminish prediction errors. However, this strategy is contingent upon the provision of ample data to the PINN for training.

(3) The impinging jet experiment reconstruction affirms the capability of PINN to accurately replicate the steady downburst wind field, maintaining precision and reliability in the presence of complex wind fields subject to topographic influences. PINN demonstrates the capacity to accurately reconstruct both the vertical and radial wind profiles pertaining to horizontal wind velocity, with errors effectively constrained to approximately 5%. However, the reconstruction accuracy for vertical wind velocity is somewhat diminished, with errors escalating to 15%.

While the PINN method has been corroborated using observational data obtained from the impinging jet experiment, further validation with comprehensive field measurement data is essential to ascertain its broad applicability. Furthermore, although PINN has substantially decreased the number of observation points for training, the quantity remains considerable for field observations of downburst.

CRediT authorship contribution statement

Binbin Yao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Zhisong Wang:** Writing – review & editing, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Zhiyuan Fang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization. **Zhengliang Li:** Writing – review & editing, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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